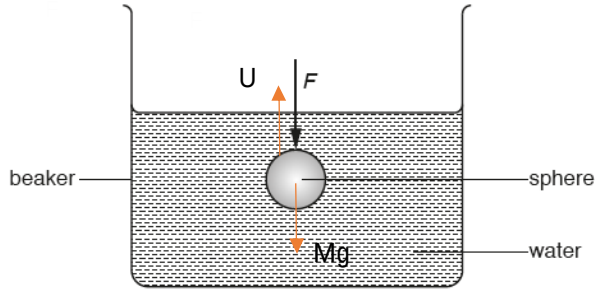


By Newton's 3rd law, magnitude of force m exerts on $3m = F = 6mg + 6ma$

6 **A** Volume of sphere = $M/\rho = 0.75/650 = 1.15 \times 10^{-3} \text{ m}^3$



For equilibrium,

$$F + Mg = U$$

$$F = U - Mg = V\rho_w g - Mg$$

$$= 1.15 \times 10^{-3} \times 10^3 \times 9.81 - 0.75 \times 9.81$$

$$= 3.9 \text{ N}$$